

TechNova Solutions Pvt. Ltd.

Enterprise Knowledge Base

Engineering Handbook

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1. Purpose

This handbook defines the engineering practices, coding standards, development workflows, and collaboration guidelines followed by TechNova Solutions Pvt. Ltd. It ensures software is developed consistently, securely, and with high quality.

2. Scope

This handbook applies to:

- Software Engineers
 - QA Engineers
 - DevOps Engineers
 - Data Engineers
 - AI/ML Engineers
 - Engineering Interns
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3. Engineering Team Structure

The Engineering Department consists of:

- Software Development
- Quality Assurance (QA)
- DevOps
- Data Engineering
- AI & Machine Learning
- Platform Engineering

Each team reports to an Engineering Manager.

4. Development Lifecycle

All software projects follow the Software Development Life Cycle (SDLC):

1. Requirement Analysis
2. Solution Design
3. Development
4. Code Review
5. Testing
6. User Acceptance Testing (UAT)
7. Production Deployment
8. Post-Deployment Monitoring

No project should skip mandatory testing or code review.

5. Coding Standards

Engineers must:

- Write clean and readable code.
- Use meaningful variable and function names.
- Follow language-specific style guides.
- Avoid duplicated code.
- Add comments only where necessary.
- Remove unused code before committing.

Code quality should be prioritized over speed.

6. Version Control (Git)

All source code is managed using Git.

Developers must:

- Commit small logical changes.
 - Write descriptive commit messages.
 - Push code regularly.
 - Never commit passwords, API keys, or secrets.
 - Resolve merge conflicts before creating a pull request.
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7. Code Review Process

Every code change requires a Pull Request (PR).

Reviewers should verify:

- Functionality
- Code quality
- Security
- Performance
- Documentation
- Test coverage

A minimum of one approved review is required before merging.

8. Branching Strategy

Standard Git branches include:

- `main` – Production-ready code
- `develop` – Active development
- `feature/*` – New features
- `bugfix/*` – Bug fixes
- `hotfix/*` – Critical production fixes
- `release/*` – Release preparation

Direct commits to the `main` branch are prohibited.

9. Build & Deployment

The CI/CD pipeline automatically:

- Builds the application
- Runs unit tests
- Executes security checks
- Creates deployment packages
- Deploys to staging
- Promotes approved builds to production

Production deployments require approval from the Release Manager.

10. Testing Standards

Every project must include:

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Regression Testing

Critical defects must be resolved before production deployment.

11. Documentation Guidelines

Every project should maintain:

- README file
- API Documentation
- Architecture Diagram
- Deployment Guide
- Troubleshooting Guide
- Release Notes

Documentation must be updated whenever significant changes are made.

12. Production Support

The Engineering Team is responsible for:

- Monitoring production applications
- Resolving incidents
- Investigating system failures
- Applying emergency fixes
- Supporting business-critical systems

All production incidents must be logged in the Incident Management System.

13. Incident Management

Incident priorities are:

Priority	Example	Target Resolution
P1	Production outage	4 Hours
P2	Major feature unavailable	8 Hours
P3	Minor bug	2 Business Days
P4	Cosmetic issue	Next planned release

Root Cause Analysis (RCA) is mandatory for all P1 incidents.

14. Development Tools

Approved tools include:

- Git
- GitHub Enterprise
- Visual Studio Code
- IntelliJ IDEA
- Docker
- Kubernetes
- PostgreSQL
- Python
- Java
- Node.js
- Jira
- Confluence

New tools require approval from the Engineering Manager.

15. Engineering Best Practices

Engineers should:

- Write reusable code.

- Automate repetitive tasks.
- Optimize application performance.
- Follow security guidelines.
- Monitor application health.
- Participate in knowledge-sharing sessions.
- Review peers' code regularly.

Continuous learning is encouraged.

16. Frequently Asked Questions

Can I push directly to the **main** branch?

No. All changes must go through a Pull Request and code review process.

How many approvals are required before merging?

At least one engineering reviewer must approve the Pull Request.

When should documentation be updated?

Documentation should be updated whenever functionality or architecture changes.

What happens if a production issue occurs?

The incident is logged, prioritized, and handled according to the Incident Management process.

Which branching strategy does TechNova use?

The company follows a Git Flow–based branching strategy using **main**, **develop**, **feature**, **bugfix**, **hotfix**, and **release** branches.

Engineering Standards Summary

Area	Standard
Version Control	Git
Code Review	Mandatory
Pull Requests	Mandatory
CI/CD	Automated
Testing	Required
Documentation	Required
Production Monitoring	Continuous

Related Documents

- Information Security Handbook
- IT Helpdesk Guide
- AI Department Handbook
- Product Documentation